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Display panel and manufacturing method thereof

Abstract

A display panel and manufacturing method thereof are provided. The display panel includes a substrate, a first electrode, a hole injection layer, a hole transport layer, an electron blocking layer, an organic electroluminescent layer, a hole blocking layer, an electron transport layer, an electron injection layer, and a second electrode arranged sequentially. The electron blocking layer corresponding to a red sub-pixel region of the organic electroluminescent layer is made of a p-type red light host material. Accordingly, the present invention effectively simplifies a manufacturing process, adjusts and balances charge carriers, and improves device performance.

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Background/Summary

RELATED APPLICATIONS

(1) This application is a National Phase of PCT Patent Application No. PCT/CN2020/082748 having International filing date of Apr. 1, 2020, which claims the benefit of priority of Chinese Patent Application No. 201911371300.X filed on Dec. 26, 2019. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE INVENTION

- (2) The present invention relates to a field of organic light-emitting displays and in particular, to a display panel and a manufacturing method thereof.
- (3) Conventional flat panel display devices include liquid crystal displays (LCDs), plasma display panels (PDP), organic light emitting diode (OLED) display devices, and etc. Especially, the OLED display devices have advantages such as low power consumption, high contrast ratios, and wide viewing angles, and being thin and flexible. OLED has now become the mainstream and next generation display technology. For OLED display devices which are active light-emitting devices, the OLED display devices include an ITO anode, a hole injection layer (HIL), a hole transport layer (HTL), an electron blocking layer (EBL), a light emitting layer (EML), a hole blocking layers (HBL), an electron transport layer (ETL), an electron injection layer (EIL), a cathode made of metal, and a capping layer (CPL). The light-emitting layer is the most important part of an OLED device, and includes a host material and a guest material. In an OLED device, electrons are injected from a cathode, and holes are injected from an anode, and an electron-hole pair is formed in a light-emitting layer to combine to generate an exciplex which is also called an exciton.
- (4) In order to improve an exciton utilization rate of the OLED device, that is, to increase the probability of energy transfer from the host material to the guest material, the following approach is taken. At present, a red light host material uses a certain ratio of an n-RH host material (n-RH) to a p-type host material (p-RH) to form a premixed material. By injecting this premixed material into electrons and holes, exciplexes are generated to increase the exciton utilization rate and thus increase efficiency and a service life of the device. However, when the red light premixed material is subjected to a evaporation deposition process, the ratio of the n-type host material to the p-type host material changes as the evaporation time goes by, resulting in low efficiency and a shorter service life of the device.
- (5) As shown in FIG. 1, in a conventional red light emitting device, a highest occupied molecular orbit (HOMO) energy level of the electron blocking layer (EBL) is poorly matched with an HOMO energy level of the p-type host material (p-RH), which causes a potential energy barrier to electron/hole migration and leads to degradation of device performance. Furthermore, the electron blocking layer and the light emitting layer need to be separately deposited in two different chambers by using at least two fine metal masks (FMM) for evaporation, so more manufacturing steps are required, and production costs are also increased.

SUMMARY OF THE INVENTION

- (6) It is an objective of the present invention to provide a display panel and a manufacturing method thereof, which only need the same one fine metal mask, and evaporation is performed in a same evaporation chamber, thus effectively simplifying a manufacturing process, adjusting and balancing charge carriers of the device, and improving device performance.
- (7) Accordingly, the present invention provides a display panel, comprising: a substrate, a first electrode, a hole injection layer, a hole transport layer, an electron blocking layer, an organic electroluminescent layer, a hole blocking layer, an electron transport layer, an electron injection layer, and a second electrode disposed sequentially, wherein the electron blocking layer corresponding to a red sub-pixel region of the organic electroluminescent layer is made of a p-type red light host material.
- (8) Preferably, the p-type red light host material is (7-(4,6-diphenylpyrimidin-2-yl)-5-phenyl-7H-benzo[c]carbazole) or (9-(3-([1,1'-biphenyl]-3-yl)quinoxalin-2-yl)-3,6-diphenyl-9H-carbazole).
- (9) Preferably, a thickness of the hole injection layer is 30 to 500 angstroms (Å), a thickness of the hole transport layer is 100 to 2000 Å, and a thickness of the electron blocking layer is 100 to 2000 Å, a thickness of the organic electroluminescent layer is 30 to 800 Å, a thickness of the hole blocking layer is 10 to 500 Å, a thickness of the electron transport layer is 50 to 800 Å, a thickness of the electron injection layer is 5 to 400 Å, and a thickness of the second electrode is 50 to 300 Å.
- (10) Preferably, the electron blocking layer further comprises a green blocking portion arranged corresponding to a green sub-pixel region of the organic electroluminescent layer, and a blue

blocking portion arranged corresponding to a blue sub-pixel region of the organic electroluminescent layer, wherein the p-type red light host material is arranged adjacent to the green blocking portion.

(11) Preferably, the display panel further comprises a cover layer, a protective layer, and an encapsulation layer sequentially disposed on the second electrode, wherein the substrate comprises a base substrate and a thin film transistor driving layer, and the first electrode is disposed on the thin film transistor driving layer.

(12) The present invention further provides a display panel, comprising a substrate, a first electrode, a hole injection layer, a hole transport layer, an electron blocking layer, an organic electroluminescent layer, a hole blocking layer, an electron transport layer, an electron injection layer, and a second electrode disposed sequentially, wherein the organic electroluminescent layer comprises a red sub-pixel region, a green sub-pixel region, and a blue sub-pixel region arranged spaced apart from each other, and the electron blocking layer corresponding to the red sub-pixel region is made of a p-type red light host material, wherein the electron blocking layer comprises a green blocking portion corresponding to the green sub-pixel region, and a blue blocking portion corresponding to the blue sub-pixel region.

(13) Preferably, the p-type red light host material is (7-(4,6-diphenylpyrimidin-2-yl)-5-phenyl-7H-benzo[c]carbazole) or (9-(3-([1,1'-biphenyl]-3-yl)quinoxalin-2-yl)-3,6-diphenyl-9H-carbazole).

(14) Preferably, a thickness of the hole injection layer is 30 to 500 angstroms (Å), a thickness of the hole transport layer is 100 to 2000 Å, and a thickness of the electron blocking layer is 100 to 2000 Å, a thickness of the organic electroluminescent layer is 30 to 800 Å, a thickness of the hole blocking layer is 10 to 500 Å, a thickness of the electron transport layer is 50 to 800 Å, a thickness of the electron injection layer is 5 to 400 Å, and a thickness of the second electrode is 50 to 300 Å.

(15) Preferably, the electron blocking layer further comprises a green blocking portion arranged corresponding to a green sub-pixel region of the organic electroluminescent layer, and a blue blocking portion arranged corresponding to a blue sub-pixel region of the organic electroluminescent layer, wherein the p-type red light host material is arranged adjacent to the green blocking portion.

(16) Preferably, the display panel further comprises a cover layer, a protective layer, and an encapsulation layer sequentially disposed on the second electrode, wherein the substrate comprises a base substrate and a thin film transistor driving layer, and the first electrode is disposed on the thin film transistor driving layer.

(17) The present invention further provides a manufacturing method of a display panel, comprising steps as follows: sequentially providing a substrate, a first electrode, a hole injection layer, a hole transport layer, an electron blocking layer, an organic electroluminescent layer, a hole blocking layer, an electron transport layer, an electron injection layer, and a second electrode; wherein the electron blocking layer corresponding to a red sub-pixel region of the organic electroluminescent layer is made of a p-type red light host material.

(18) Preferably, when the p-type red light host material is prepared in the red sub-pixel region, a baffle of a p-type red light host material evaporation source is opened, and a baffle of a red light dopant material evaporation source and a baffle of an n-type red light host material evaporation source are closed, so that the p-type red light host material is deposited on the hole transport layer.

(19) Preferably, when the organic electroluminescent layer is produced, the baffles of the p-type red light host material evaporation source, the red light dopant material evaporation source, and the n-type red light host material evaporation source are opened at the same time, and the p-type red light host material, the red light dopant material, and the n-type red light host material are mixed together.

(20) Preferably, a mixing ratio of the p-type red light host material and the n-type red light host material is adjusted by an evaporation rate, and a doping ratio of the red light dopant material is adjusted by the evaporation rate.

(21) Preferably, areas for evaporation deposition of the p-type red light host material, the red light dopant material, and the n-type red light host material are controlled by an angle limiting plate of a fine metal mask.

(22) The present invention has the following advantages. In the present invention, a p-type red light host material is used to replace a material of a portion of the electron blocking layer corresponding to the red sub-pixel region of the organic electroluminescent layer. Evaporation equipment requires no extra evaporation chambers to produce the electron blocking layer, so a manufacturing process is simplified, and manufacturing costs are reduced. Furthermore, the mixing ratio of the p-type red light host material (p-RH) and the n-type red light host material (n-RH) in the display panel of the present invention is effectively controlled by the evaporation rate through the evaporation equipment, and no changes occur as the evaporation time passes, thereby increasing the quality and reliability of the device.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) In order to more clearly illustrate the embodiments of the present disclosure or related art, figures which will be described in the embodiments are briefly introduced hereinafter. It is obvious that the drawings are merely for the purposes of illustrating some embodiments of the present disclosure, and a person having ordinary skill in this field can obtain other figures according to these figures without inventive work.

(2) FIG. 1 is a schematic diagram illustrating energy levels of a conventional display panel.

(3) FIG. 2 is a schematic cross-sectional view illustrating a display panel according to the present invention.

(4) FIG. 3 is a schematic cross-sectional view illustrating an electron blocking layer and an organic electroluminescent layer of the display panel according to the present invention.

(5) FIG. 4 is a schematic view illustrating evaporation deposition for the display panel in an evaporation chamber of evaporation equipment.

(6) FIG. 5 is another schematic view illustrating evaporation deposition for the display panel in the evaporation chamber of the evaporation equipment.

(7) FIG. 6 is a schematic view illustrating energy levels of the display panel of the present invention.

(8) FIG. 7 is a process flow diagram illustrating a manufacturing method of the display panel according to the present invention.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

(9) In the present disclosure, “embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one embodiment of the present invention. The use of the same terms in different places in the specification is not necessarily limited to the same embodiment, but should be understood as being for independent or alternative embodiments. Based on the technical solutions disclosed in the embodiments of the present invention, those skilled in the art should be able to make modifications or combinations according to the concept of the present invention.

(10) Please refer to FIGS. 2 and 3. FIG. 2 is a schematic cross-sectional view illustrating a display panel of the present invention. FIG. 3 is a schematic cross-sectional view illustrating an electron blocking layer and an organic electroluminescent layer of the display panel of the present invention. As shown in the drawings, the present invention provides a display panel 1 including a substrate 101, a first electrode 1011, a hole injection layer 102, a hole transport layer 103, an electron blocking layer 104, an organic electroluminescent layer 105, a hole blocking layer 106, an electron transport layer 107, an electron injection layer 108, and a second electrode 109. The

electron blocking layer **104** corresponding to a red sub-pixel region **1051** of the organic electroluminescent layer **105** is made of a p-type red light host material **1041**.

(11) The display panel **1** can be used in, for example, a wearable device such as a smart bracelet, a smart watch, a virtual reality (VR) device, a mobile phone, an e-book/e-newspaper, a television, a personal portable computer, a foldable/flexible OLED display device. In the present embodiment, the p-type red light host material **1041** is (7-(4,6-diphenylpyrimidin-2-yl)-5-phenyl-7H-benzo[c]carbazole), and its bond-line formula is as follows:

(12) ##STR00001##

(13) Alternatively, the p-type red light host material **1041** is (9-(3-([1,1'-biphenyl]-3-yl)quinoxalin-2-yl)-3,6-diphenyl-9H-carbazole), and its bond-line formula is as follows:

(14) ##STR00002##

(15) The substrate **101** comprises, but is not limited to, a rigid substrate or a flexible substrate, and its structure includes a base substrate (not illustrated) and a thin film transistor (TFT) driving layer (not illustrated) arranged sequentially. The first electrode **1011** is disposed on the TFT driving layer, and the first electrode **1011** is an anode (generally, indium tin oxide, or ITO for short). The TFT driving layer comprises, but is not limited to, an a-Si TFT, a low temperature poly-silicon (LTPS) TFT, or other suitable TFTs. The second electrode **109** is a cathode and has a thickness of 50-300 angstroms (Å). When the second electrode **109** is electrically connected to the first electrode **1011**, sub-pixels in the organic electroluminescent layer **105** are driven to emit light.

(16) As shown in FIG. 3, the electron blocking layer **104** includes a green blocking portion **1042** corresponding to the green sub-pixel region **1052** of the organic electroluminescent layer **105**, and also includes a blue blocking portion **1043** corresponding to the blue sub-pixel region **1053** of the organic electroluminescent layer **105**. The p-type red light host material **1041** is arranged adjacent to the green blocking portion **1042**. In a preferable embodiment, the display panel **1** can emit only red light; however, in other embodiments, the display panel **1** can also include green sub-pixels (not labeled) and blue sub-pixels (not labeled) and emit visible light of various colors.

(17) The hole injection layer **102** is disposed on the substrate **101**. The hole injection layer **102** is, for example, a conventional p-type dopant material, a Copper Phthalocyanine (CuPc), titanyl phthalocyanine (TiOPc), m-MTDATA, 2-TNATA, or other energy level matching materials. A thickness of the hole injection layer **102** is 30 to 500 Å. The hole transport layer **103** is disposed on the hole injection layer **102** and has a thickness of 100 to 2000 Å. The electron blocking layer **104** is disposed on the hole transport layer **103** and has a thickness of 100 to 2000 Å.

(18) The hole blocking layer **106** is disposed on the organic electroluminescent layer **105** and has a thickness of 10 to 500 Å. The electron transport layer **107** is disposed on the hole blocking layer **106** and has a thickness of 50 to 800 Å. Material of the electron transport layer **107** includes, but is not limited to, one or more of Alq3, Almq3, DVPBi, TAZ, OXD, PBD, BND, and PV. The electron injection layer **108** is disposed on the electron transport layer **107** and has a thickness of 5 to 400 Å.

(19) In the embodiment shown in FIG. 2, the present invention further includes a cover layer **110**, a protective layer **111**, and an encapsulation layer **112** disposed on the second electrode **109** in sequence. The cover layer **110** is disposed on the second electrode **109** and has a thickness of 20 to 2000 Å. The cover layer **110** is a film layer for improving a light output rate. Material of the protective layer **111** includes, but is not limited to, lithium fluoride (LiF). The encapsulation layer **112** simultaneously covers the protective layer **111**, the cover layer **110**, the second electrode **109**, the electron injection layer **108**, the electron transport layer **107**, the hole blocking layer **106**, the organic electroluminescent layer **105**, the electron blocking layer **104**, the hole transport layer **103**, the hole injection layer **102**, and the first electrode **101** to protect the display panel **1** from being damaged by impurities such as moisture and dust.

(20) The present invention further provides a manufacturing method of a display panel. As shown in FIG. 7, the manufacturing method includes following steps. Step S1: sequentially providing a

substrate **101**, a first electrode **1011**, a hole injection layer **102**, a hole transport layer **103**, an electron blocking layer **104**, an organic electroluminescent layer **105**, a hole blocking layer **106**, an electron transport layer **107**, an electron injection layer **108**, and a second electrode **109**. A portion of the electron blocking layer **104** corresponding to a red sub-pixel region **1051** of the organic electroluminescent layer **105** is made of a p-type red light host material **1041**.

(21) Please refer to FIGS. **4** to **6**. Vacuum multi-effect evaporation equipment **2** is taken as an example for description. When evaporation-depositing the p-type red light host material **1041** corresponding to the red sub-pixel region **1051** of the organic electroluminescent layer **105** is performed in a same evaporation chamber **21** of the evaporation equipment **2**. A p-type red light host material evaporation source **22**, a red light dopant material evaporation source **23**, and an n-type red light host material evaporation source **24** are respectively connected to the evaporation chamber **21** at intervals. When the p-type red light host material **1041** is evaporated, a baffle (not illustrated) of the p-type red light host material evaporation source **22** is opened, and a baffle **232** of the red light dopant material evaporation source **23** and a baffle **242** of the n-type red light host material evaporation source **24** are closed, so that a p-type red light host material **221** (organic dye) is uniformly and stably evaporation-deposited on the hole transport layer **103**.

(22) The organic electroluminescent layer **105** is produced by mixing and depositing the p-type red host material (p-RH) **221**, the red light dopant material (RD) **231**, and the n-type red host material (n-RH) **241**. A thickness of the organic electroluminescent layer **105** ranges from 30 Å to 800 Å. When the organic electroluminescent layer **105** is deposited, the baffle (not illustrated) of the p-type red light host material evaporation source **22** and the baffle **232** of the red light dopant material evaporation source **23** and the baffle **242** of the n-type red light host material evaporation source **24** are opened at the same time, so that the p-type red light host material **221**, the red light dopant material **231**, and the n-type red light host material **241** are mixed in the evaporation chamber **21**.

(23) A mixing ratio of the p-type red light host material **221** and the n-type red light host material **241** is adjusted by an evaporation rate through the evaporation equipment **2**, and a doping ratio of the red light dopant material **231** is also adjusted by the evaporation rate. Areas for evaporation deposition of the p-type red light host material **221**, the red light dopant material **231**, and the n-type red light host material **241** are controlled by an angle limiting plate (not illustrated) of a fine metal mask (not illustrated).

(24) In another embodiment, before depositing the p-type red light host material **1041**, the same fine metal mask is used to deposit the electron blocking layer and a blue light-emitting material (not illustrated), and the same fine metal mask is used to deposit the electron blocking layer and a red light-emitting material (not illustrated) on the blue light-emitting material; configuration may vary as needed to produce display devices as required. Please also refer to FIG. **6**. When the p-type red light host material **1041** (p-RH) of the present embodiment is used to replace a portion of the electron blocking layer (EBL) **104** corresponding to the red sub-pixel region **1051**, the p-type red light host material **1041** has a highest occupied molecular orbital (HOMO) energy level that matches a HOMO energy level of the red sub-pixel region **1051** in the organic electroluminescent layer **105**, thus enhancing electron  custom character // hole  custom character migration, thereby balancing charge carriers of the display panel **1** and improving product performance thereof.

(25) The present embodiment further includes the protective layer **111** deposited on the cover layer **110** and the encapsulation layer **112** deposited on the protective layer **111**. The encapsulation layer **112** covers the hole injection layer **102**, the hole transport layer **103**, the electron blocking layer **104**, the organic electroluminescent layer **105**, the hole blocking layer **106**, the electron transport layer **107**, the electron injection layer **108**, a cathode **109**, the cover layer **110** on the substrate **101** by physical vapor deposition (PVD), chemical vapor deposition (CVD) or other suitable deposition methods, so that the display panel **1** is encapsulated and protected from the influence of moisture and oxygen, and the device performance is not reduced.

(26) Methods of producing the layers on the substrate **101** include, but are not limited to, evaporation deposition, inkjet printing, or other suitable methods. The specific method is described in the foregoing examples, and will not be repeated here.

(27) It should be noted that, depositing the p-type red light host material **1041** and the organic electroluminescent layer **105** are carried out using the same fine metal mask (not illustrated) in the same evaporation chamber **21**, so the manufacturing process is effectively simplified, the charge carriers of the device are balanced, and the device performance is improved. Specifically, in the present embodiment, the p-type red light host material **1041** is used to replace a material of a portion of the electron blocking layer **104** corresponding to the red sub-pixel region **1051** of the organic electroluminescent layer **105**. Therefore, it is not necessary to prepare an additional evaporation chamber (not illustrated) in the equipment and a different fine metal mask (not illustrated) for producing the electron blocking layer **104**. Accordingly, a manufacturing process is simplified, and production costs are reduced. Furthermore, a mixing ratio of the p-type red light host material **221** (p-RH) and the n-type red light host material **241** (n-RH) in the display panel **1** is effectively controlled by an evaporation rate of the evaporation equipment **2**, and does not change as the evaporation time increases, thereby increasing device quality and reliability.

(28) In summary, although the present invention has been described in connection with specific embodiments, it should be understood that many alternatives, modifications, and variations will be apparent to those skilled in the art. The present invention is therefore intended to cover all alternatives, modifications, and variations that fall within the scope of the appended claims.

Claims

1. A display panel, comprising a substrate, a first electrode, a hole injection layer, a hole transport layer, an electron blocking layer, an organic electroluminescence layer, a hole blocking layer, an electron transport layer, an electron injection layer, and a second electrode disposed sequentially, wherein the organic electroluminescent layer comprises a red sub-pixel region, a green sub-pixel region, and a blue sub-pixel region arranged spaced apart from each other, the organic electroluminescent layer comprises a p-type red light host material and a n-type red light host material, and the electron blocking layer corresponding to the red sub-pixel region of the organic electroluminescent layer is made of the p-type red light host material, wherein a portion of the electron blocking layer corresponding to the red sub-pixel region, and the red sub-pixel region of the organic electroluminescent layer, comprise the same p-type red light host material, wherein the p-type red light host material in the portion of the electron blocking layer corresponding to the red sub-pixel region exhibits a highest occupied molecular orbital (HOMO) energy level that matches the HOMO energy level of the red sub-pixel region in the organic electroluminescent layer.
2. The display panel according to claim 1, wherein the p-type red light host material is (7-(4,6-diphenylpyrimidin-2-yl)-5-phenyl-7H-benzo[c]carbazole) or (9-(3-([1,1'-biphenyl]-3-yl)quinoxalin-2-yl)-3,6-diphenyl-9H-carbazole).
3. The display panel according to claim 1, wherein a thickness of the hole injection layer is 30 to 500 angstroms (Å), a thickness of the hole transport layer is 100 to 2000 Å, and a thickness of the electron blocking layer is 100 to 2000 Å, a thickness of the organic electroluminescent layer is 30 to 800 Å, a thickness of the hole blocking layer is 10 to 500 Å, a thickness of the electron transport layer is 50 to 800 Å, a thickness of the electron injection layer is 5 to 400 Å, and a thickness of the second electrode is 50 to 300 Å.
4. The display panel according to claim 1, wherein the electron blocking layer further comprises a green blocking portion arranged corresponding to a green sub-pixel region of the organic electroluminescent layer, and a blue blocking portion arranged corresponding to a blue sub-pixel region of the organic electroluminescent layer, wherein the p-type red light host material is arranged adjacent to the green blocking portion.

5. The display panel according to claim 1, further comprising a cover layer, a protective layer, and an encapsulation layer sequentially disposed on the second electrode, wherein the substrate comprises a base substrate and a thin film transistor driving layer, and the first electrode is disposed on the thin film transistor driving layer.
6. A display panel, comprising a substrate, a first electrode, a hole injection layer, a hole transport layer, an electron blocking layer, an organic electroluminescent layer, a hole blocking layer, an electron transport layer, an electron injection layer, and a second electrode disposed sequentially, wherein the organic electroluminescent layer comprises a red sub-pixel region, a green sub-pixel region, and a blue sub-pixel region arranged spaced apart from each other, the organic electroluminescent layer comprises a p-type red light host material and a n-type red light host material, and the electron blocking layer corresponding to the red sub-pixel region is made of the p-type red light host material, wherein the electron blocking layer comprises a green blocking portion corresponding to the green sub-pixel region, and a blue blocking portion corresponding to the blue sub-pixel region, wherein a portion of the electron blocking layer corresponding to the red sub-pixel region, and the red sub-pixel region of the organic electroluminescent layer, comprise the same p-type red light host material, wherein the p-type red light host material in the portion of the electron blocking layer corresponding to the red sub-pixel region exhibits a highest occupied molecular orbital (HOMO) energy level that matches the HOMO energy level of the red sub-pixel region in the organic electroluminescent layer.
7. The display panel according to claim 6, wherein the p-type red light host material is (7-(4,6-diphenylpyrimidin-2-yl)-5-phenyl-7H-benzo[c]carbazole) or (9-(3-([1,1'-biphenyl]-3-yl)quinoxalin-2-yl)-3,6-diphenyl-9H-carbazole), a thickness of the electron blocking layer is 100 to 2000 angstroms (Å), and a thickness of the organic electroluminescent layer is 30 to 800 Å.
8. The display panel according to claim 6, wherein a thickness of the hole injection layer is 30 to 500 Å, a thickness of the hole transport layer is 100 to 2000 Å, a thickness of the electron blocking layer is 100 to 2000 Å, a thickness of the organic electroluminescent layer is 30 to 800 Å, a thickness of the hole blocking layer is 10 to 500 Å, a thickness of the electron transport layer is 50 to 800 Å, a thickness of the electron injection layer is 5 to 400 Å, and a thickness of the second electrode is 50 to 300 Å.
9. The display panel according to claim 6, wherein the electron blocking layer further comprises a green blocking portion corresponding to a green sub-pixel region of the organic electroluminescent layer, a blue blocking portion corresponding to a blue sub-pixel region of the organic electroluminescent layer, and the p-type red light host material is arranged adjacent to the green blocking portion.
10. The display panel according to claim 6, further comprising a cover layer, a protective layer, and an encapsulation layer sequentially disposed on the second electrode, the substrate comprising a base substrate and a thin film transistor driving layer, wherein the first electrode is disposed on the thin film transistor driving layer.
11. A manufacturing method of a display panel, comprising steps as follows: sequentially providing a substrate, a first electrode, a hole injection layer, a hole transport layer, an electron blocking layer, an organic electroluminescent layer, a hole blocking layer, an electron transport layer, an electron injection layer, and a second electrode; wherein the organic electroluminescent layer comprises a red sub-pixel region, a green sub-pixel region, and a blue sub-pixel region arranged spaced apart from each other, the organic electroluminescent layer comprises a p-type red light host material and a n-type red light host material, and the electron blocking layer corresponding to the red sub-pixel region of the organic electroluminescent layer is made of the p-type red light host material, wherein a portion of the electron blocking layer corresponding to the red sub-pixel region, and the red sub-pixel region of the organic electroluminescent layer, comprise the same p-type red light host material, wherein the p-type red light host material in the portion of the electron blocking layer corresponding to the red sub-pixel region exhibits a highest occupied molecular orbital (HOMO)

energy level that matches the HOMO energy level of the red sub-pixel region in the organic electroluminescent layer.

12. The manufacturing method of the display panel according to claim 11, wherein when the p-type red light host material is prepared in the red sub-pixel region, a baffle of a p-type red light host material evaporation source is opened, and a baffle of a red light dopant material evaporation source and a baffle of an n-type red light host material evaporation source are closed, so that the p-type red light host material is deposited on the hole transport layer.

13. The manufacturing method of the display panel according to claim 12, wherein when the organic electroluminescent layer is produced, the baffles of the p-type red light host material evaporation source, the red light dopant material evaporation source, and the n-type red light host material evaporation source are opened at the same time, and the p-type red light host material, the red light dopant material, and the n-type red light host material are mixed together.

14. The manufacturing method of the display panel according to claim 12, wherein a mixing ratio of the p-type red light host material and the n-type red light host material is adjusted by an evaporation rate, and a doping ratio of the red light dopant material is adjusted by the evaporation rate.

15. The manufacturing method of the display panel according to claim 12, wherein areas for evaporation deposition of the p-type red light host material, the red light dopant material, and the n-type red light host material are controlled by an angle limiting plate of a fine metal mask.
